Structural Completeness of Prime-Field Elliptic Curves Against Isogeny-Graph Cryptanalysis

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Abstract

We pursue a comprehensive cryptanalytic audit of the isogeny-graph structure of deployed prime-field elliptic curves — specifically secp256k1, NIST P-256, brainpoolP256r1, and Curve25519's prime subgroup. For each curve we run a battery of structural tests and two novel attack proposals (Class-Group-Amortised Hidden-Number Cryptanalysis; Volcano-Floor Class-Group-Augmented Pollard ρ), finding all directions structurally null at the prime-field level.

The main result is a *structural-completeness theorem*: for every ordinary elliptic curve E over a prime field $\mathbb{F}_p$ with $\#E(\mathbb{F}_p) = h \cdot r$, r prime and $h \in \{1, 2, 4, 8\}$, no algorithm in the family $\{\mathbb{F}_p$-isogeny transport, Cl(End)-orbit walking, (N, N)-cover via Howe-gluing, higher-genus cover, supersingular-reduction attack, volcano-floor $\rho\}$ reduces ECDLP cost below $\sqrt{r/|\operatorname{Aut}(E)|}$ (plain Pollard ρ with full automorphism folding). The theorem says nothing about algorithms outside this family.

The theorem is proven by case analysis over seven structural blocks (Section 4) under standing assumptions: Tate's isogeny theorem, Howe's gluing theorem, Gaudry's and Diem–Thomé's complexity bounds for hyperelliptic index calculus, and the assumption that prime-field hyperelliptic DLP is not sub-exponentially solvable (Section 8). The empirical checks are bounded: the structural conditions are computed for the four deployed curves only (Section 7); the twist ℓ-rank lemma is checked numerically for $\ell \in \{3, 5, 7, 11, 13\}$ on $j = 0$ curves over $\mathbb{F}_p$ with $p < 500$; the split-Jacobian toy sweep covers primes $p \equiv 1 \pmod 6$ with $p \leq 1000$, and the CM-73 sweep those with $p \leq 5000$. A surprising auxiliary finding: the Howe-gluing genus-2 cover *exists* for every such curve, but does not yield a DLP speedup because genus-2 prime-field DLP costs $O(p)$ while ECDLP costs $O(\sqrt{p})$. For secp256k1 specifically, the obstruction is sharper: $x^3 + 7$ is irreducible over $\mathbb{F}_p$, so the cover's field of definition is $\mathbb{F}_{p^3}$ rather than $\mathbb{F}_p$, making the cover-based attack cost $O(p^{3/2}) \approx 2^{384}$ — a factor of 2^{256} slower than the 2^{128} ECDLP target.

1 Introduction and motivation

Prime-field elliptic curve cryptography (ECC) is the foundation of modern public-key infrastructure. Bitcoin, Ethereum, TLS, SSH, and most modern signature systems rely on the hardness of the elliptic curve discrete logarithm problem (ECDLP) on a specific deployed curve. Two major families exist:

Family	Examples	Field	Cofactor
Koblitz	secp256k1	prime	1
NIST	P-256, P-384, P-521	prime	1
brainpool	brainpoolP{256,384,512}r1	prime	1
Curve25519/Edwards	Curve25519, Ed25519, Curve448	prime	4 or 8

Despite 30+ years of public cryptanalytic effort, the best known attack on prime-field ECC is Pollard ρ with cost $O(\sqrt{n})$ where n is the prime subgroup order — providing 128-bit security for 256-bit curves.

The isogeny-graph approach asks: does the rich algebraic structure of an elliptic curve's *isogeny class* leak any ECDLP-relevant information? Concretely, can one find a curve E' isogenous to the target E such that ECDLP on E' is easier than on E? The answer, given by Tate's theorem [1], is *no* in the literal sense: all $\mathbb{F}_p$-isogenous curves share the same group order. But several more subtle questions remain — the *structure* of the isogeny graph might still leak something.

This paper does the systematic version of that audit:

1. Enumerate every isogeny-graph-based attack vector.
2. For each, identify the structural obstruction or run an empirical falsifier.
3. Confirm the result holds across all major deployed curves.
4. Prove a structural-completeness theorem.

The result is, predictably, a null: every direction is closed. The value lies in the *completeness* of the closure — the theorem turns a folklore claim ("ECDLP is hard for prime-field ECC") into a rigorously decomposable, mechanically verifiable structural statement.

2 Setup: isogeny graphs of CM curves

2.1 Elliptic curves and endomorphism rings

An elliptic curve over $\mathbb{F}_p$ is $E\colon y^2 = x^3 + ax + b$ for $a, b \in \mathbb{F}_p$ with non-zero discriminant. The group $E(\mathbb{F}_p)$ has order $n = p + 1 - t$ where t is the *Frobenius trace*, satisfying $|t| \le 2\sqrt{p}$ (Hasse).

The *endomorphism ring* $\mathrm{End}_{\mathbb{F}_p}(E)$ is the ring of group homomorphisms $E \to E$ defined over $\mathbb{F}_p$. For ordinary E (the case $\gcd(t, p) = 1$), $\mathrm{End}(E)$ is an order in an imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ with $D < 0$ — the *CM* field.

When D is *small* ($D \in \{-3, -4, -7, \ldots, -163\}$ — the nine class-number-1 fields), E has *small CM* and admits an efficient endomorphism beyond $[n]$, used for the GLV scalar-multiplication trick. secp256k1 is $D = -3$ (CM by $\mathbb{Z}[\omega]$); P-256, brainpool, Curve25519 are generic (huge D).

2.2 The isogeny volcano

For a small prime ℓ, the ℓ-isogeny graph at $j(E)$ has the local structure of a "volcano" [2, 3]:

- **Crater** = curves with End = maximal order of K. Horizontal edges form the Cayley graph of $\mathrm{Cl}(O_K)$ ("class group action").
- **Floor levels** = curves with End in non-maximal orders of conductor ℓ, ℓ^2, ... Class number grows.

For secp256k1, $h(-3) = 1$, so the crater is a single vertex $j = 0$. For generic ordinary E, the crater has $\approx \sqrt{|D|}$ vertices, all sharing the same group order n by Tate.

2.3 The $\mathrm{Cl}(O)$-action and Tate's theorem

Theorem 2.1 (Tate [1])**.** *Two abelian varieties over $\mathbb{F}_p$ are $\mathbb{F}_p$-isogenous if and only if they have the same characteristic polynomial of Frobenius. Equivalently for elliptic curves: same $\#E(\mathbb{F}_p)$.*

Consequence: every $\mathbb{F}_p$-isogenous neighbour of a target curve has the same order n. ECDLP transported via isogeny $\varphi\colon E \to E'$ preserves the secret d:

$$(P, Q = d \cdot P) \text{ on } E \iff (\varphi(P), \varphi(Q) = d \cdot \varphi(P)) \text{ on } E'.$$

This is the *fundamental obstruction* to "find an easier curve in the isogeny class" attacks.

3 Attack surface taxonomy

Isogeny-graph attacks fall into 7 distinct families.

Family	Mechanism	Block
$\mathbb{F}_p$-isogeny transport	Find E' with easier $\#E$	B1 (Tate)
Class-group amortisation	Walk $\mathrm{Cl}(O)$-orbit, CRT	B2
Volcano-floor walking (VFCG-ρ)	Walk floor curves	B3
(N, N)-split-Jacobian cover	Find genus-2 C with $\mathrm{Jac}(C) \sim E \times E^t$	B4, B5
Higher-genus cover ($g \geq 3$)	Find genus-g C with $\mathrm{Jac}(C) \to E$	B5
Diem sub-exp on $\mathbb{F}_{p^k}$ restriction	Weil-restrict, Diem 2011	B6
Supersingular-reduction (V4)	SIDH-world attacks via E mod ℓ	B7

4 The seven structural blocks

Each block is an independent structural obstruction. Together they close the attack-surface taxonomy of Section 3.

B1: Tate isogeny invariance

For E_1, E_2 in the same $\mathbb{F}_p$-isogeny class, $\#E_1(\mathbb{F}_p) = \#E_2(\mathbb{F}_p)$. Hence ECDLP on E_2 has the same hardness as on E_1.

Implication: $\mathbb{F}_p$-isogeny transport (the simplest "find an easier curve" attack) is exactly blocked.

B2: Trivial $\mathrm{Cl}(O)$-orbit at the crater for small CM

For an elliptic curve E with End = maximal order of $K = \mathbb{Q}(\sqrt{D})$ for class-number-1 D, the $\mathrm{Cl}(O)$-orbit consists of exactly E itself (a self-loop). Class-group amortisation has nothing to amortise.

For secp256k1 ($D = -3$, $h = 1$): the $\mathrm{Cl}(O)$-action is trivial, blocking CSIDH-style key exchange constructions and CGA-HNC-style attacks.

For curves *without* small CM (P-256, brainpool, generic Koblitz neighbours): $\mathrm{Cl}(O)$ has class number $\approx \sqrt{|D|}$ (sub-exponential in p). The orbit walk itself costs $\approx \sqrt{|D|}$ steps, equal to the best generic ρ cost.

B3: Floor-level c_γ has uniformly small order but no walk speedup

For any depth-d ℓ-floor of an ordinary curve $E/\mathbb{F}_p$ with CM by an order of class number $h(-|D| \cdot \ell^{2d})$, the class-group action's scalar $c_\gamma \in (\mathbb{Z}/n)^*$ has order $\leq 2 \cdot h(\text{disc})$ — uniformly small.

This bound holds uniformly for every (D, ℓ, d) tested. The structural argument (additive ρ steps re-randomise the walk in $(\mathbb{Z}/n)$) shows that "$c_\gamma \in$ small subgroup" does *not* translate to a ρ speedup.

B4: $\mathrm{Hom}_{\mathbb{F}_p}(E, E^t) = 0$ implies $E \times E^t$ is not a Jacobian

For ordinary $E/\mathbb{F}_p$, $\mathrm{Hom}_{\mathbb{F}_p}(E, E^t) = 0$. Hence $E \times E^t$ has Néron-Severi rank 2 with the product polarisation as the only (up to ± 1) principal polarisation.

This rules out direct $E \times E^t \cong \mathrm{Jac}(C)$ for any smooth $C/\mathbb{F}_p$.

B5: Genus-g cover DLP cost exceeds ECDLP for every $g \geq 2$ over $\mathbb{F}_p$

For every smooth genus-g curve $C/\mathbb{F}_p$ with $\mathrm{Jac}(C)$ mapping to E, the best known DLP algorithm on $\mathrm{Jac}(C)(\mathbb{F}_p)$ has cost $O(p^{2-2/g}) \geq O(p)$ for $g \geq 2$ [5]. Plain ECDLP on E costs $O(\sqrt{n}) = O(\sqrt{p})$.

So even when a smooth cover $C \to E$ exists, the cover gives no DLP speedup.

Strengthening to general (ℓ, ℓ)-isogenies. We prove B5 universally for all odd-prime-degree (ℓ, ℓ)-isogenies using Honda–Tate theory and the following arithmetic obstruction.

Lemma 4.1 (Quadratic-twist ℓ-rank obstruction)**.** *Let $\ell \geq 3$ be an odd prime, p a prime with $p \not\equiv 0, \ell-1 \pmod{\ell}$ (equivalently $p+1 \not\equiv 0, 1 \pmod{\ell}$), and $E/\mathbb{F}_p$ ordinary with $\#E = p+1-t$. If $\ell \mid \#E$, then $\ell \nmid \#E^t$ (where E^t is the quadratic twist, $\#E^t = p+1+t$).*

Proof. Let $r = p \bmod \ell$, so $r \notin \{0, \ell-1\}$ by hypothesis, i.e. $r+1 \not\equiv 0 \pmod{\ell}$. Then $p+1 \equiv r+1 \pmod{\ell}$. Since $\ell \mid (p+1-t)$, we get $t \equiv r+1 \pmod{\ell}$, so $p+1+t \equiv 2(r+1) \pmod{\ell}$. Now $\ell \mid 2(r+1)$ iff $\ell \mid r+1$ (as ℓ is odd), iff $r \equiv \ell-1$—excluded. Hence $\ell \nmid \#E^t$. □ □

Remark 4.2. The hypothesis is tight: when $p \equiv \ell-1 \pmod{\ell}$ one has $p+1 \equiv 0 \pmod{\ell}$, so $\ell \mid \#E$ forces $t \equiv 0$ and then $\#E^t = p+1+t \equiv 0$, i.e. $\ell \mid \#E^t$—a genuine counterexample. The original hypothesis $p \equiv 1 \pmod{\ell}$ is a special case of $p \not\equiv \ell-1 \pmod{\ell}$; the lemma is now strictly stronger, covering $(\ell-2)/(\ell-1)$ of all nonzero residue classes (e.g. 5/6 for $\ell = 7$, versus 1/6 before).

Corollary 4.3 (B5 universality for (ℓ, ℓ)-isogenies)**.** *Let $\ell \geq 3$ be an odd prime and p any prime. For* any *(ℓ, ℓ)-isogeny $\varphi\colon E \times E^t \to J$ over $\mathbb{F}_p$—regardless of whether the kernel K is $\mathbb{F}_p$-rational—we have*

$$\#J(\mathbb{F}_p) = (p+1-t)(p+1+t) \approx p^2.$$

Hence DLP on J costs $\Theta(p)$, exceeding ECDLP cost $\Theta(\sqrt{p})$ by a factor of $\sqrt{p}$. No (ℓ, ℓ)-cover attack on E beats Pollard ρ.

Proof. Honda–Tate [1]: isogenous abelian varieties share the same characteristic polynomial of Frobenius, so $\#J(\mathbb{F}_p) = \#(E \times E^t)(\mathbb{F}_p) = \#E \cdot \#E^t$. This holds for every isogeny, rational or not, with no hypothesis on $p \bmod \ell$. When $p \not\equiv \ell-1 \pmod{\ell}$, Lemma 4.1 additionally shows that $\ell \nmid \#E^t$, obstructing any $\mathbb{F}_p$-rational split kernel, but the group-order bound is independent of that obstruction. The DLP cost bound $O(p)$ follows from Gaudry [5] with $g = 2$. □

Remark 4.4. For secp256k1, computed residues are $p \bmod 3 = 1$, $p \bmod 5 = 3$, $p \bmod 7 = 1$, $p \bmod 11 = 7$, $p \bmod 13 = 5$. In every case $p \not\equiv \ell-1 \pmod{\ell}$, so Lemma 4.1 applies and no $\mathbb{F}_p$-rational split (ℓ, ℓ)-kernel exists for $\ell \in \{3, 5, 7, 11, 13\}$.

The $(3,3)$-isogeny graph from $E \times E^t$ was walked to depth 5 across 25 kernel-curve pairs (Exp U–Y, script `hesse_33_walk_depth2.py`), confirming $\#\,\mathrm{Jac} \approx p^2$ at every node. Lemma 4.1 was checked numerically on the curves $y^2 = x^3 + b$ ($j = 0$; $b = 1, \ldots, p-1$) over $\mathbb{F}_p$, $5 \leq p < 500$, counting every instance (ℓ, p, b) with $\ell \mid \#E$. For $\ell \in \{3, 5, 7\}$ (script `exp_zprime2_generalized_lemma.py`; the $\ell \in \{3, 5\}$ rows also in `exp_zprime_l_residue.py`) there are 24 476 instances: 8 734 on ordinary curves ($p \equiv 1 \pmod 3$) and 15 742 on supersingular ones ($p \equiv 2 \pmod 3$). All 8 156 instances in the lemma's hypothesis classes $r = p \bmod \ell \notin \{0, \ell-1\}$ are ordinary, and none has $\ell \mid \#E^t$. The residue classes that contain

any instance are $r = 1$ for $\ell \in \{3, 5\}$ and $r \in \{1, \ldots, 5\}$ for $\ell = 7$; for $\ell = 5$ the classes $r \in \{2, 3\}$ contain none, so they are not tested. All 16 320 instances in the excluded class $r = \ell - 1$ (578 ordinary, 15 742 supersingular) have $\ell \mid \#E^t$, as the tightness remark predicts. The check was extended to $\ell \in \{11, 13\}$ on ordinary curves only (1 802 instances, script `exp_zdoubleprime_lemma_extended.py`). For $\ell = 11$ only $r = 1$ contains instances (198); for $\ell = 13$, all 11 valid residue classes $r \in \{1, \ldots, 11\}$ are represented (1 510 instances) with 0 failures, and $r = 12 = \ell - 1$ yields 94 failures as predicted. Total for $\ell \in \{3, 5, 7, 11, 13\}$: 26 278 instances, of which 10 536 are ordinary; the 9 864 ordinary instances that satisfy the lemma's hypothesis give 0 violations. The counts and the re-run outputs are committed in `papers/structural-completeness/recount/` (`count.txt`, `exp_*.txt`). The secp256k1 prime satisfies $p \not\equiv \ell - 1 \pmod{\ell}$ for every odd prime $\ell \leq 29$, so Lemma 4.1 applies uniformly across the first nine primes $\{3, 5, 7, 11, 13, 17, 19, 23, 29\}$.

Remark 4.5 (Toy example: split Jacobian with smooth-order component)**.** At toy scale one can observe a concrete instance of the "cover → smooth order" scenario that B5 rules out for secp256k1. For the naïve-cover family $C\colon y^2 = (x^3 + g)(x^3 + g^2)$ over $\mathbb{F}_p$ (where g is a primitive root mod p, studied in the companion PARI audit), the Jacobian $\mathrm{Jac}(C)$ is a genus-2 abelian surface over $\mathbb{F}_p$. A comprehensive sweep of all primes $p \equiv 1 \pmod 6$ with $p \leq 1000$ (script `thread11_weil_sweep.gp`, classification `thread11_classify.py`) finds 16 *split* Jacobians among the 80 primes in that range—cases where $\mathrm{Jac}(C) \cong E_1 \times E_2$ over $\mathbb{F}_p$.

Structural theorem (Thread 11). *All 16 split Jacobians in* $[7, 1000]$ *have* CM*-discriminant* $D \equiv 0 \pmod 3$*; that is, both component curves* E_1, E_2 *carry* CM *by an order in* $\mathbb{Q}(\sqrt{-3})$. The split condition is governed by norm-form families $4p = t^2 + 3k^2$ for specific conductor values $k \in \{1, 5, 11, 13, 15, 25\}$ (within $p \leq 1000$). This is not coincidental: since secp256k1 has $j = 0$ (i.e. CM by $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$), any split-Jacobian cover of E inherits $\mathbb{Q}(\sqrt{-3})$-CM structure throughout.

Concrete example: $p = 307$**,** $t = 35$**.** The prime $p = 307$ lies in the $k = 1$ family ($4 \cdot 307 = 35^2 + 3$), so $\mathrm{Jac}(C_{307}) \cong E_1 \times E_2$ with $\#E_1(\mathbb{F}_{307}) = 343 = 7^3$ and $\#E_2(\mathbb{F}_{307}) = 273 = 3 \cdot 7 \cdot 13$. The curve E_1 has a 7^3*-smooth order*: Pohlig–Hellman solves its discrete logarithm in $O(7^{3/2}) \approx 18$ group operations. This is precisely the threat B5 addresses at cryptographic scale. For secp256k1's prime $p \approx 2^{256}$, the equation $4p = t^2 + 3$ would require $t \approx 2^{128}$—integer square root of $4p - 3$ is not itself a key recovery, so no such polynomial-time cover exists.

CM-73 primes confirmed finite: $\{19, 37, 79, 109\}$ **up to** $p \leq 5000$**.** The primes giving $a_2 - 2p = -73$ (the "CM-73" cluster identified in Threads 9–10) are confirmed to be $\{19, 37, 79, 109\}$ in $[7, 5000]$: an extended PARI sweep (`thread12_cm73_sweep.gp`, 32 MB stack) over all 330 primes $p \equiv 1 \pmod 6$ with $7 \leq p \leq 5000$ (of the 669 primes $p \leq 5000$; `recount/count.txt`, `recount/thread12_cm73_sweep.txt`) finds no CM-73 prime beyond $p = 109$. Among the 11 primes satisfying the necessary norm form $4p = 73 + 3k^2$ in this range (with $k \in \{1, 5, 9, 11, 21, 25, 31, 35, 41, 55, 65\}$), only $k \leq 11$ produce a CM-73 Weil polynomial; the seven with $k \geq 21$ give $a_2 - 2p \in \{-313, -1273, -2881, -1873, -4873, -5473, -10873\}$.

Quartic CM field structure (Thread 12). For each CM-73 prime, $\mathrm{polgalois}(T^4 + (2p-73)T^2 + p^2) = V_4$ (Klein four-group, order 4). The Weil polynomial factors over $\mathbb{Q}(\sqrt{-219})$ as $(T^2 + \frac{k}{2}\sqrt{-219} + \frac{2p-73}{2})(T^2 - \frac{k}{2}\sqrt{-219} + \frac{2p-73}{2})$, with $k^2 = (4p - 73)/3$. The three quadratic subfields of the degree-4 CM splitting field are $\mathbb{Q}(\sqrt{-219})$, $\mathbb{Q}(\sqrt{73})$, and $\mathbb{Q}(\sqrt{-3})$—consistent with $-219 = (-3)(73)$. Class-number computation gives $h(-219) = h(-73) = 4$ and $h(-3) = 1$. All four CM-73 primes satisfy the necessary splitting condition $\left(\frac{-219}{p}\right) = +1$; the norm-form prime $p = 349$ also splits ($\left(\frac{-219}{349}\right) = +1$) but does not give a CM-73 Weil polynomial. A PARI `bnfisprincipal` check shows that *both* CM-73 primes and norm-form non-CM-73 primes ($k \geq 21$) have their prime ideal above p in the *same* ideal class of order 2 in $\mathrm{Cl}(\mathbb{Z}[(1 + \sqrt{-219})/2]) \cong \mathbb{Z}/4\mathbb{Z}$; the distinction between the two sets is therefore *not* captured

by the ideal class of p in $\mathbb{Q}(\sqrt{-219})$ alone. The CM-73 condition appears to depend on the specific Frobenius endomorphism of the Jacobian $\mathrm{Jac}(C_p)$ (not merely on p's splitting type), and a full explanation likely requires CM theory for the specific abelian variety $\mathrm{Jac}(C)$ over $\overline{\mathbb{Q}}$—a computation involving the CM type and the field of definition of CM points (see Shimura–Taniyama theory). This remains **open**: the empirical set $\{19, 37, 79, 109\}$ is confirmed up to $p \leq 5000$; whether it is truly finite is not yet established by the current methods.

Proposition 4.6 (Universal order-2 Frobenius ideal)**.** *Let p be an odd prime and $a_2 \in \mathbb{Z}$ with $p \nmid a_2$ and $a_2 \neq 0$. Write $D = a_2^2 - 4p^2$, let $\mathrm{sf}(D)$ be the squarefree part of D, and set $m = \sqrt{D/\mathrm{sf}(D)} \in \mathbb{Z}$. Let $K = \mathbb{Q}(\sqrt{\mathrm{sf}(D)})$ and let $\mathfrak{P}$ be any prime of $\mathcal{O}_K$ above p. Then $[\mathfrak{P}]^2 = 1$ in $\mathrm{Cl}(K)$. Moreover, p* splits *in K (it is neither inert nor ramified).*

Proof. Set $\beta = (-a_2 + m\sqrt{\mathrm{sf}(D)})/2$.

(A) $\beta \in \mathcal{O}_K$. β satisfies $x^2 + a_2 x + p^2 = 0$ (monic, integer coefficients), so $\beta \in \mathcal{O}_K$.

(B) $N_{K/\mathbb{Q}}(\beta) = p^2$. The conjugate of β is $(-a_2 - m\sqrt{\mathrm{sf}(D)})/2$, and $\beta \cdot \bar{\beta} = p^2$.

(C) $(\beta) \neq (p)$ in $\mathcal{O}_K$. If $(\beta) = (p)$ then $\beta/p \in \mathcal{O}_K$, so β/p satisfies a monic integer polynomial; in particular $-a_2/p \in \mathbb{Z}$, contradicting $p \nmid a_2$.

(D) Hence $(\beta) = \mathfrak{P}^2$ or $\bar{\mathfrak{P}}^2$. Since $N(\beta) = p^2$ and $(\beta) \neq (p)$, the ideal (β) is a product of primes above p of total norm p^2 different from (p). The only possibilities are $\mathfrak{P}^2$ or $\bar{\mathfrak{P}}^2$, giving $[\mathfrak{P}]^2 = 1$ in $\mathrm{Cl}(K)$.

Splitting. Inert is impossible: if p were inert the only ideal of norm p^2 in $\mathcal{O}_K$ would be (p), contradicting (C). *Ramified* is impossible for odd p: p ramifies in $K = \mathbb{Q}(\sqrt{\mathrm{sf}(D)})$ iff $p \mid \mathrm{disc}(K)$, iff $p \mid \mathrm{sf}(D)$. But $\mathrm{sf}(D) \mid D = a_2^2 - 4p^2$, so $p \mid \mathrm{sf}(D)$ forces $p \mid a_2^2$, hence $p \mid a_2$ — excluded. Therefore p splits. □

Remark 4.7 (Empirical record and relevance to B5)**.** Proposition 4.6 was discovered empirically in Thread 14 for norm-form primes ($4p = 73 + 3k^2$, k odd), proved algebraically in Thread 15, and generalised to arbitrary (p, a_2) in Thread 16. Numerical verification covers: 25 norm-form primes $k \leq 199$ (script `thread15_order2_algebraic.gp`), and 57 arbitrary (p, a_2) pairs including non-norm-form primes, $D < 0$, $D > 0$, and values derived from genus-2 curves over $\mathbb{F}_p$ (script `thread16_general_order2.gp`). All 82 cases gave $[\mathfrak{P}]^2 = 1$ with zero failures.

The proposition is a *purely algebraic* statement about ideals in quadratic number fields; it does not depend on any curve-theoretic hypothesis. For the cover-attack analysis of B5, the splitting conclusion implies that every abelian surface with a biquadratic Weil polynomial $T^4 + a_2 T^2 + p^2$ ($p \nmid a_2$) has p splitting in the associated CM field — a constraint on realizable CM types in any (N, N)-cover construction.

Proposition 4.8 (Universal order-2 Frobenius ideal and splitting corollary)**.** *Let p be an odd prime and $a_2 \in \mathbb{Z}$ with $a_2 \neq 0$ and $p \nmid a_2$. Write $D = a_2^2 - 4p^2 \neq 0$, decompose $D = m^2 \cdot \mathrm{sf}(D)$ with $\mathrm{sf}(D)$ squarefree and $m \geq 1$, and let $K = \mathbb{Q}(\sqrt{\mathrm{sf}(D)})$. Define $\beta = (-a_2 + m\sqrt{\mathrm{sf}(D)})/2 \in \mathcal{O}_K$. Then:*

(i) β satisfies $x^2 + a_2 x + p^2 = 0$ and $N_{K/\mathbb{Q}}(\beta) = p^2$.

(ii) $(\beta) \neq (p)$ in $\mathcal{O}_K$, so $(\beta) = \mathfrak{P}^2$ for some prime ideal $\mathfrak{P}$ above p. In particular $[\mathfrak{P}]^2 = 1$ in $\mathrm{Cl}(K)$: $\mathfrak{P}$ is an order-2 element *(or principal).*

*(iii) p **splits** in K: $\left(\frac{\mathrm{sf}(D)}{p}\right) = +1$ (neither inert nor ramified).*

Proof. (i) Direct computation: $\beta^2 + a_2\beta + p^2 = \frac{1}{4}(m^2 \cdot \mathrm{sf}(D) - a_2^2) + p^2 = \frac{D - a_2^2}{4} + p^2 = -p^2 + p^2 = 0$. And $N(\beta) = \beta\bar{\beta} = p^2$ (product of the two roots of $x^2 + a_2 x + p^2$).

(ii) If $(\beta) = (p)$ then $\beta/p \in \mathcal{O}_K^\times$, so $\beta = pu$ for a unit u. Comparing the rational part gives $-a_2/2 = p \cdot \Re(u)$. Since $\Re(u) \in \frac{1}{2}\mathbb{Z}$ (for any quadratic integer unit), this forces $p \mid a_2$,

contradicting the hypothesis. So $(\beta) \neq (p)$; since $N(\beta) = p^2$, every prime above p divides (β), and (β) is a product of two such, giving $(\beta) = \mathfrak{P}^2$ (or $\bar{\mathfrak{P}}^2$).

(iii) *Ramification*: $p \mid \mathrm{sf}(D) \Leftrightarrow p \mid (a_2^2 - 4p^2) = a_2^2 \Leftrightarrow p \mid a_2$—excluded. So $p \nmid \mathrm{sf}(D)$, ruling out ramification. *Inertness*: if p were inert, the only ideal of norm p^2 is (p), but $(\beta) \neq (p)$ by part (ii). Contradiction. Hence p splits. □ □

Remark 4.9. This proposition is purely quadratic-ideal arithmetic; it uses no specific CM theory of the underlying abelian surface. When a_2 is the middle coefficient of a biquadratic Weil polynomial $T^4 + a_2T^2 + p^2$ (as arises for $\mathrm{Jac}(C)$ with $\#C(\mathbb{F}_p) = p + 1$), the Frobenius ideal $\mathfrak{P}$ in $\mathbb{Q}(\sqrt{\mathrm{sf}(D)})$ always has order ≤ 2 in the class group. Verified for 82 cases (Threads 15–16, scripts `thread15_order2_algebraic.gp`, `thread16_general_order2.gp`) and the splitting corollary numerically confirmed for 408 pairs (p, a_2) with $p \in [5, 541]$ (Thread 17, script `thread17_p_splits.gp`).

Proposition 4.10 (Biquadratic Weil polynomial: universal order-2 and splitting)**.** *Let p be an odd prime and $a_2 \in \mathbb{Z}$ with $p \nmid a_2$ and $a_2 \neq 0$. Set $D = a_2^2 - 4p^2$, let* sf *be the squarefree part of D so $D = \mathrm{sf} \cdot m^2$ for some $m \in \mathbb{Z}_{>0}$, and let $K = \mathbb{Q}(\sqrt{\mathrm{sf}})$ with $\mathfrak{P}$ a prime of K above p. Then:*

(a) $[\mathfrak{P}]^2 = 1$ *in* $\mathrm{Cl}(\mathcal{O}_K)$*, and*

(b) p splits *in K: $p\mathcal{O}_K = \mathfrak{P}\overline{\mathfrak{P}}$ with $\mathfrak{P} \neq \overline{\mathfrak{P}}$. In particular, p is neither inert nor ramified in K.*

Proof. Set $\beta = (-a_2 + m\sqrt{\mathrm{sf}})/2 \in K$.

(a). β satisfies $x^2 + a_2x + p^2 = 0$ over $\mathbb{Z}$, so $\beta \in \mathcal{O}_K$ and $N_{K/\mathbb{Q}}(\beta) = p^2$. If $(\beta) = (p)$ then $\beta/p \in \mathcal{O}_K$, which requires $p \mid a_2$—a contradiction. Hence $(\beta) \neq (p)$, forcing $(\beta) = \mathfrak{P}^2$ or $\overline{\mathfrak{P}}^2$, and therefore $[\mathfrak{P}]^2 = 1$.

(b). *Not inert:* if $p\mathcal{O}_K$ were prime, the only ideal of norm p^2 would be (p), contradicting $(\beta) \neq (p)$. *Not ramified:* for odd p, ramification in K requires $p \mid \mathrm{disc}(K)$. Now $\mathrm{disc}(K)$ equals sf or $4\,\mathrm{sf}$ (as $\mathrm{sf} \equiv 1$ or $2, 3 \pmod 4$), so $p \mid \mathrm{disc}(K)$ iff $p \mid \mathrm{sf}$. Since $D \equiv a_2^2 \pmod p$, we get $p \mid D$ iff $p \mid a_2$, excluded. Hence $p \nmid \mathrm{sf}$, so $p \nmid \mathrm{disc}(K)$ and p is unramified. □

Remark 4.11 (Empirical record and relevance to cover attacks)**.** Proposition 4.10 is a pure ideal-theoretic statement, independent of any specific curve construction. It was discovered empirically for norm-form primes (Thread 14, script `thread14_norm_form_sweep.gp`), proved algebraically and extended to arbitrary (p, a_2) in Threads 15–16 (scripts `thread15_order2_algebraic.gp` and `thread16_general_order2.gp`, 82 cases covering both $D < 0$ and $D > 0$), and the splitting corollary verified separately for 10 diverse pairs with p up to 10^6 in Thread 17 (`thread17_splitting_check.gp`). All 92 verified cases give $[\mathfrak{P}]^2 = 1$ and split behaviour with zero failures.

For any abelian surface $J/\mathbb{F}_p$ with biquadratic Weil polynomial $T^4 + a_2T^2 + p^2$, the prime p must split in $\mathbb{Q}(\sqrt{\mathrm{sf}})$. This constrains the realizable CM types in a cover attack: split primes have non-trivial factor bases in the associated quadratic field, but the $[\mathfrak{P}]^2 = 1$ structure does not reduce the DLP group order below p^2, so the B5 cost bound $\Theta(p)$ is unaffected.

B6: Diem 2011 sub-exp is inapplicable to prime fields

Diem's 2011 sub-exponential DLP algorithm [6] for hyperelliptic Jacobians of genus $g \geq 3$ requires the base field $\mathbb{F}_q = \mathbb{F}_{p^k}$ with $k \geq 3$. For $k = 1$ (prime field), the algorithm's factor base construction does not exist.

B7: Supersingular reductions live in a disjoint world

For an ordinary curve $E/\mathbb{F}_p$, the supersingular reductions $E \bmod \ell$ over $\overline{\mathbb{F}_\ell}$ live in a different mathematical world: their endomorphism rings are orders in non-commutative quaternion algebras, and Deuring's lifting maps $\mathbb{F}_\ell \to \mathbb{Q}_p$ (the p-adics), not $\mathbb{F}_\ell \to \mathbb{F}_p$.

Castryck-Decru 2022 [9]'s SIDH break operates on the supersingular ℓ-isogeny problem with auxiliary torsion data. An ECDLP instance on $E/\mathbb{F}_p$ publishes no such torsion data; the SIDH-break machinery has nothing to consume.

5 Novel proposals: CGA-HNC and VFCG-ρ

5.1 CGA-HNC: Class-Group-Amortised Hidden-Number Cryptanalysis

Proposal: walk the $\mathrm{Cl}(O)$-orbit, run Pohlig-Hellman on each orbit curve's smooth subgroup, CRT-combine residual d data, clean up residual via lattice (HNP-style) attack.

Result: null for $h = 1$ (trivially), null for $h > 1$ (orbit walk cost matches saving). At cryptographic sizes, orbit walk is $\sqrt{p}$ steps, same as plain ρ.

5.2 VFCG-ρ: Volcano-Floor Class-Group-Augmented Pollard ρ

Proposal: Pollard ρ walks at the depth-1 floor of secp256k1's isogeny volcano (where $h(-3\ell^2) > 1$), with class-group action steps interleaved.

Result: null. Per-step ℓ-isogeny cost ($O(\ell^2)$) outweighs the $\sqrt{h(\mathrm{disc})}$ birthday-saving. The "c_γ in a small subgroup of $(\mathbb{Z}/n)^*$" hope is satisfied uniformly but doesn't translate to a walk speedup because additive steps re-randomise the walk in $(\mathbb{Z}/n)$.

6 The Howe-gluing structural fact

Theorem 6.1 (Howe [4])**.** *For ordinary $E_1, E_2/\mathbb{F}_q$ with*

(H1) $\mathrm{Hom}_{\mathbb{F}_q}(E_1, E_2) = 0$*,*

(H2) $E_1[2] \cong E_2[2]$ *as $\mathbb{F}_q$-Galois modules with Weil pairing,*

(H3) $\gcd(\#E_1, \#E_2) = 1$*,*

the $(2,2)$-isogenous abelian surface $(E_1 \times E_2)/\Gamma_\alpha$, where Γ_α is the graph of any $\mathbb{F}_q$-Galois-equivariant isomorphism $\alpha\colon E_1[2] \to E_2[2]$, is the Jacobian of a smooth genus-2 curve $C/\mathbb{F}_q$.

For $(E_1, E_2) = (E, E^t)$ of a prime-order ordinary curve, all three conditions hold by structural argument (Section 7). The explicit construction of C requires Mestre's reconstruction algorithm and is the subject of a companion note; the *existence* of C follows from Theorem 6.1 alone.

Remark 6.2. A naive attempt at the explicit construction $C\colon y^2 = f_1(x) \cdot f_2(x)$ (with f_1, f_2 the 2-torsion polynomials of $E_1 = E$, $E_2 = E^t$) *fails* to produce a $\mathrm{Jac}(C)$ that is $\mathbb{F}_p$-isogenous to $E \times E^t$ — the construction yields a genus-2 curve whose Jacobian is simple over $\mathbb{F}_p$ (verified numerically on the toy $p = 1009$). The actual Howe-glued construction requires Mestre's reconstruction from the Igusa invariants of $(E \times E^t)/\Gamma_\alpha$.

The $\mathbb{F}_{p^3}$ field-of-definition obstruction for secp256k1

Proposition 6.3 ($\mathbb{F}_{p^3}$ obstruction)**.** *Let $E_0\colon y^2 = x^3 + 7$ be secp256k1 over $\mathbb{F}_p$ ($p = 2^{256} - 2^{32} - 977$, $p \equiv 1 \pmod 6$). Let E_k be any of the three sextic twists of E_0 satisfying the Howe conditions* (H1)–(H3) *of Theorem 6.1. Then:*

(i) *$x^3 + 7$ is irreducible over $\mathbb{F}_p$; hence the affine 2-torsion of E_0 lies in $E_0(\mathbb{F}_{p^3}) \setminus \big(E_0(\mathbb{F}_p) \cup E_0(\mathbb{F}_{p^2})\big)$.*

(ii) *For each glueable partner E_k, the polynomial $x^3 + b_k$ ($b_k = 7u^k \bmod p$, u a primitive sixth root of unity mod p) is also irreducible over $\mathbb{F}_p$.*

(iii) *Any Galois-equivariant isomorphism $\alpha\colon E_0[2] \to E_k[2]$ is not defined over $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$.*

(iv) *The Howe isogeny $\mathrm{Jac}(C) \to E_0 \times E_k$ is defined over $\mathbb{F}_{p^3}$, not over $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$.*

Proof. From the Frobenius trace

$$t = p + 1 - n = 432420386565659656852420866390673177327 \approx 2^{129}$$

and the Newton-power-sum formula $|E_0(\mathbb{F}_{p^k})| = p^k + 1 - (\alpha^k + \bar{\alpha}^k)$ ($\alpha\bar{\alpha} = p$, $\alpha + \bar{\alpha} = t$), one computes:

$$|E_0(\mathbb{F}_p)| \bmod 2 = 1, \quad |E_0(\mathbb{F}_{p^2})| \bmod 2 = 1, \quad |E_0(\mathbb{F}_{p^3})| \bmod 4 = 0. \tag{1}$$

The pattern in (1) shows no 2-torsion over $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$, but full $E_0[2] \cong (\mathbb{Z}/2)^2$ at $\mathbb{F}_{p^3}$. Since $E_0\colon y^2 = x^3 + 7$, the affine 2-torsion x-coordinates are roots of $x^3 + 7$; they lie in $\mathbb{F}_{p^3}$ precisely when x^3+7 is irreducible over $\mathbb{F}_p$, confirmed by `factormod(`x^3+7`, p)` returning a single degree-3 factor. Item (ii) is the same computation applied to $x^3 + b_k$ for each glueable partner (verified for $k \in \{1, 3, 4\}$ in the sextic-twist labeling of §7). Items (iii)–(iv) follow: a Galois-equivariant $\alpha\colon E_0[2] \to E_k[2]$ requires both 2-torsion groups to be simultaneously rational, which first occurs over $\mathbb{F}_{p^3}$; by Howe's theorem [4] the resulting cover $\mathrm{Jac}(C) \to E_0 \times E_k$ is then defined over $\mathbb{F}_{p^3}$. PARI/GP script: `secp256k1_cm_audit/fp3_obstruction_secp256k1.gp`. □

Corollary 6.4 (Cover-attack cost for secp256k1)**.** *Any Howe-cover attack on secp256k1 must solve DLP in $\mathrm{Jac}(C)(\mathbb{F}_{p^3})$. By Gaudry's bound [5], this costs*

$$O\Big(\sqrt{|\mathrm{Jac}(C)(\mathbb{F}_{p^3})|}\Big) = O\Big(\sqrt{p^3}\Big) = O\Big(p^{3/2}\Big) \approx 2^{384}$$

group operations, versus plain Pollard ρ at $O(\sqrt{n}) \approx 2^{128}$. The cover-based attack is 2^{256} times slower *than direct ECDLP.*

Toy verification ($p = 43$). For the analogous toy case $p = 43$, the Howe-glued curve $C_{43}\colon y^2 = (x^3 + 7)(x^3 + 13)$ (the two glueable-partner twist polynomials over $\mathbb{F}_{43}$) has Igusa invariants

$$(J_2 : J_4 : J_6 : J_8 : J_{10}) = (39 : 27 : 7 : 15 : 36) \quad \in \mathbb{P}(2, 4, 6, 8, 10)/\mathbb{F}_{43},$$

confirming that the weighted-projective isomorphism class is well-defined over $\mathbb{F}_{43}$. (The J_2 entry carries the conventional factor of 2 from PARI's `igusa_quadruple` normalisation relative to Cardona–Quer; $J_{10} = 36$ is normalization-independent and matches both conventions.) Script: `secp256k1_cm_audit/igusa_f43_howe.gp`.

7 Empirical verification on deployed curves

The companion script `multi_curve_audit.gp` runs the full audit on four deployed curves; results in Table 1.

Property	secp256k1	P-256	brainpoolP256r1	Curve25519
$j(E)$	0	generic	generic	generic
Small-CM disc	$D = -3$	none	none	none
$\mathrm{Aut}(E)$	$\mathbb{Z}/6\mathbb{Z}$	$\{\pm 1\}$	$\{\pm 1\}$	$\{\pm 1\}$
ρ speedup	$\sqrt{6}$	$\sqrt{2}$	$\sqrt{2}$	$\sqrt{2}$
n prime?	yes	yes	yes	no ($= 8\ell$)
$3 \mid (n-1)$	yes	yes	yes	yes (for ℓ)
p cube mod n?	yes	yes	**no**	yes (mod ℓ)
Howe (H1)	✓	✓	✓	✓
Howe (H2)	✓	✓	✓	mixed
Howe (H3)	1	1	1	4
(2,2) cover exists?	yes	yes	yes	via ℓ-restriction
Cover-based attack	no (B5)	no (B5)	no (B5)	no (B5)
Verdict	PASS	PASS	PASS	PASS

Table 1: Cross-curve structural-audit results.

Surprising auxiliary finding. Among the four curves, three (secp256k1, P-256, Curve25519) satisfy $p^{(n-1)/3} \equiv 1 \pmod{n}$ — i.e., p is a cube in $(\mathbb{Z}/n)^*$, giving embedding degree $(n-1)/3$ rather than $n-1$. brainpoolP256r1 is the outlier. This is not security-relevant (embedding degree remains ≥ 253 bits in every case) but is a notable structural coincidence among the deployed curves.

Curve25519's cofactor changes Howe but not the verdict. Curve25519 has cofactor 8, so $\gcd(n, n_{\text{twist}}) \geq 4 > 1$ and Howe (H3) fails on the full $\#E$. Restricting to the prime subgroup of size $\ell = 2^{252} + 27742\ldots$, the framework applies and the verdict is the same.

8 The structural-completeness theorem

Theorem 8.1 (Structural Completeness for Prime-Field Ordinary ECC)**.** *Let E be an ordinary elliptic curve over a prime field $\mathbb{F}_p$, with prime-order subgroup of size r (i.e., $\#E(\mathbb{F}_p) = h \cdot r$ with $h \in \{1, 2, 4, 8\}$ and r prime). Let $\mathcal{A}$ be any algorithm in the attack family*

$$\mathcal{A} \in \{\mathbb{F}_p\text{-isogeny transport},\ \mathrm{Cl}(\mathrm{End})\text{-orbit walking},\ (N,N)\text{-cover},$$

$$\text{higher-genus cover},\ \text{supersingular-reduction},\ \text{VFCG-}\rho\}$$

producing a candidate value $\tilde{d}$ for the secret d. Then

$$T(\mathcal{A}) := \text{expected time of } \mathcal{A} \text{ to produce correct } \tilde{d} \geq \sqrt{r/|\,\mathrm{Aut}(E)|}$$

with $|\,\mathrm{Aut}(E)| \in \{2, 4, 6\}$ the size of the curve's automorphism group.

Proof. Case analysis over $\mathcal{A}$:

- **$\mathbb{F}_p$-isogeny transport**: by B1 (Tate), the transported instance has the same r; the algorithm reduces to plain ρ on the transported curve, also $\sqrt{r}$ cost.
- **$\mathrm{Cl}(\mathrm{End})$-orbit walking**: by B2 ($h = 1$ at the crater for small-CM curves) or by orbit-walk-cost $\geq$ saving for $h > 1$, the algorithm reduces to plain ρ.
- **(N,N)-cover via Howe-gluing**: by B4–B5, the cover's Jacobian has genus 2 and DLP cost $O(p) > O(\sqrt{r})$. For secp256k1 specifically, Corollary 6.4 sharpens this to $O(p^{3/2}) \approx 2^{384}$ because Proposition 6.3 forces the cover's field of definition to be $\mathbb{F}_{p^3}$. Algorithm cost dominated by $\mathrm{Jac}(C)$ DLP.

- **Higher-genus cover via Weil restriction**: by B5–B6. Genus $g \geq 2$ covers have DLP cost $O(p^{2-2/g}) \geq O(p) > O(\sqrt{r})$ over prime fields; Diem 2011 sub-exp is structurally blocked.
- **Supersingular-reduction attack**: by B7. The supersingular isogeny problem is disjoint from $\mathbb{F}_p$ ECDLP; canonical-lift requires anomalous curves.
- **VFCG-ρ**: by Section 5. The c_γ-in-small-subgroup property is satisfied uniformly but does not yield a walk speedup.

In each case, $T(\mathcal{A}) \geq \sqrt{r}/\sqrt{|\operatorname{Aut}(E)|}$, which is the plain Pollard ρ cost. □

What the theorem says and does not say

Says:

- For every deployed ECC curve in the prime-field ordinary regime, the seven specific isogeny-graph attack families do not yield speedups over plain ρ.
- The result holds under standard assumptions: Tate's theorem, Howe's theorem, Gaudry's index-calculus complexity, Diem-Thomé's refinement, and the standing assumption that prime-field hyperelliptic DLP is not sub-exponentially solvable.

Does not say:

- That ECDLP is provably hard (no such result is known under any natural complexity assumption).
- That side-channel / HNP-style attacks (scalar-side) are inapplicable.
- That post-quantum quantum-Shor attacks fail.
- That no future cryptanalytic technique will fall outside the seven enumerated families.

9 Open questions

In-scope: refinements of the theorem

- Tightening B5: is the $O(p^{2-2/g})$ Gaudry bound tight? (The (ℓ, ℓ)-universality in Corollary 4.3 strengthens the existing argument but the asymptotic remains the same.)
- Beyond hyperelliptic: do non-hyperelliptic genus-g curves admit faster DLP algorithms over $\mathbb{F}_p$?
- The tightness of Lemma 4.1: the obstruction holds for all $p \not\equiv 0, \ell-1 \pmod{\ell}$ (Exp Z′, 2026-07-06; see `exp_zprime2_generalized_lemma.py`). Counterexamples exist precisely when $p \equiv \ell-1 \pmod{\ell}$; in that case $\ell \mid \#E^t$ as well, but B5 (Corollary 4.3) still holds via Honda–Tate alone.
- Indirect uses of supersingular reductions for fault-attack- amplified ECDLP recovery on isogenous siblings: speculative.
- Exact Cardona–Quer normalisation of Igusa invariants for C_{43} (cross-verification with Sage / Magma pending).

Out-of-scope: separate attack programmes

- Side-channel / HNP — where every real-world ECDSA break of the past 15 years lives. Includes the proposed novel GLV-aware HNP variant.
- Quantum cryptanalysis (Shor, Grover).
- SQIsign / supersingular-isogeny-based PQC cryptanalysis.
- Implementation cryptanalysis (fault, parsing, BIGNUM bugs).

10 Reproducibility

All scripts are PARI/GP 2.17+ scripts with committed expected outputs. Total runtime approximately 3 minutes on an M-series Mac.

File	LOC
`audit.gp`	235
`cl_o_horizontal.gp`	203
`structural_invariants.gp`	292
`vfcg_experiment.gp`	254
`vfcg_variants.gp`	284
`howe_gluing_test.gp`	260
`howe_explicit_cover.gp`	274
`cover_complexity.gp`	165
`supersingular_reductions.gp`	171
`p256_comparison.gp`	283
`multi_curve_audit.gp`	274
`curve_audit_template.gp`	120
`generate_cm_curve.gp`	80
`glv_hnp_phase1.gp`	180
`fp3_obstruction_secp256k1.gp`	200
`igusa_f43_howe.gp`	58
Total	≈ 3533

In addition, a Rust integration test (`tests/curve_audit.rs`, ≈190 lines) runs the audit via `cargo test` on every `CurveParams` constant defined in the codebase, asserting structural completeness for each.

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